

48V FRESHJET X

9620018598



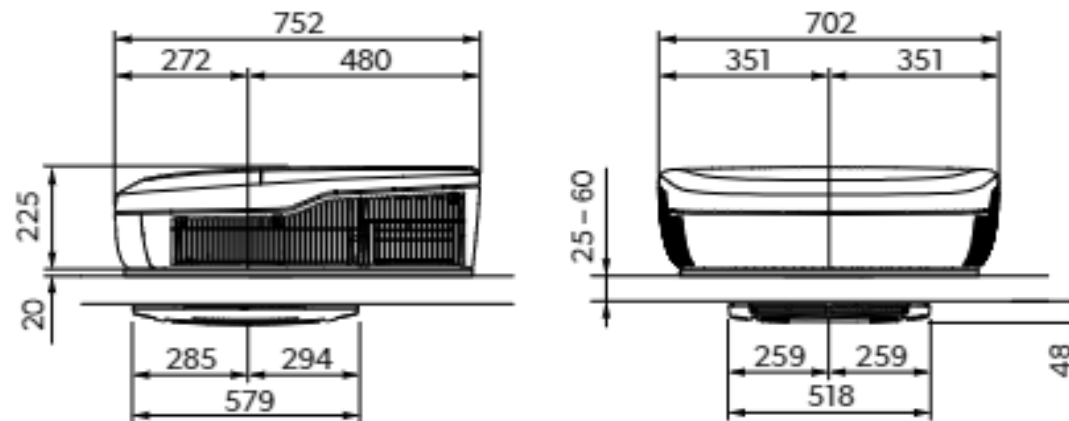
TOPICS

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SPECIFICATIONS

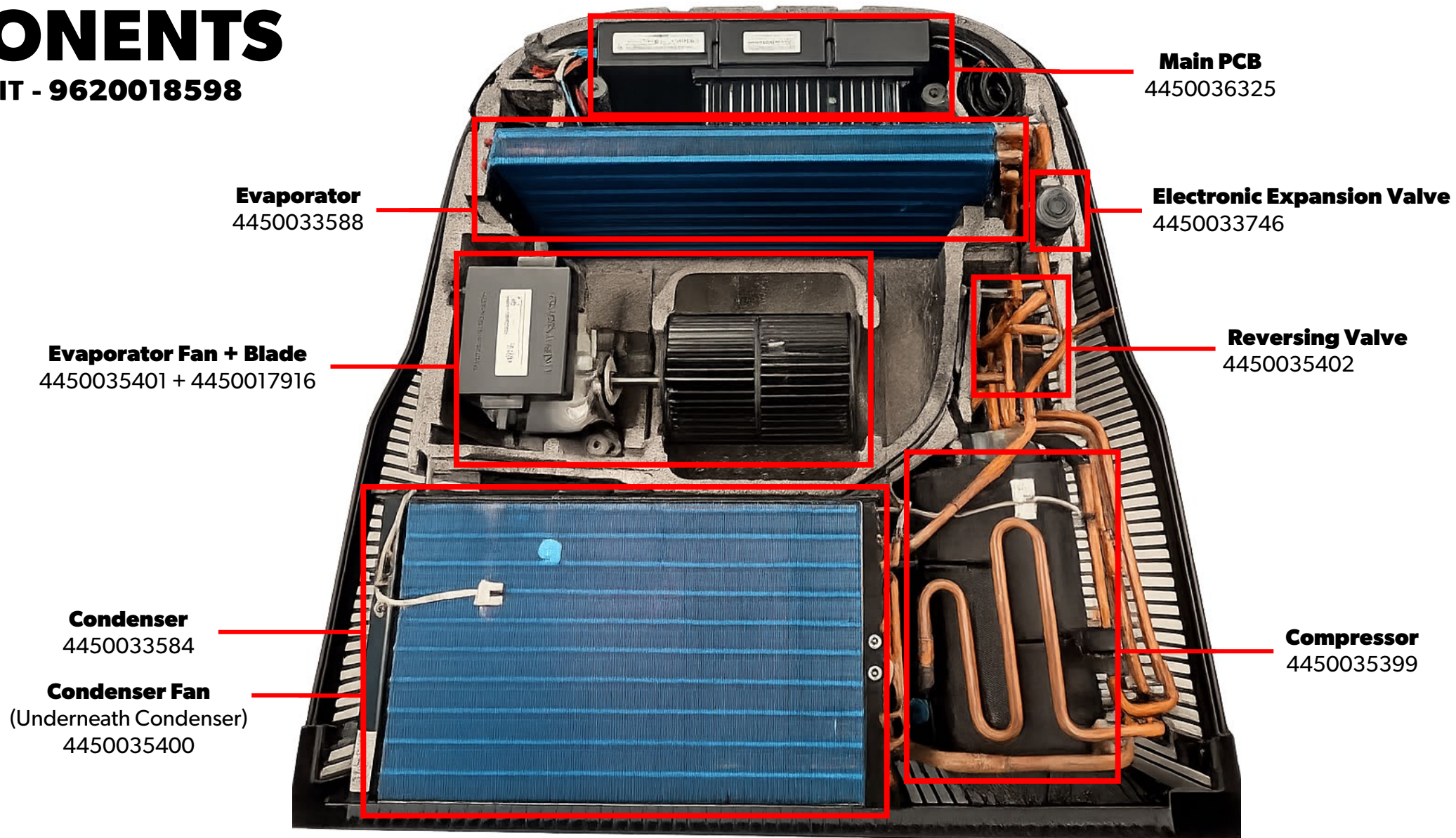
	FJZ 48V
SKU	9620018598
Invertor	Yes
Cooling Capacity	2523 W
Heating Capacity	2218 W
DC Voltage Range	45 – 51 V
Max Current Cooling	< 28.6 A
Max Current Heating	< 20.3 A
Operation Temperature Range	-2°C to 52°C
Refrigerant	R32 (420g)
Weight	35kg
Dimensions L x W x H	752mm x 702mm x 225mm
Colour	Grey
Caravan/RV sizing (Recommendation for optimal performance)	13ft – 19ft



Recommended Caravan sizing subject to RV/Caravan build quality, insulation properties, furnishings and design.

COMPONENTS

ROOFTOP UNIT - 9620018598



COMPONENTS

ADB - 9620001693

(The same ADB as the 240v models)



Remote Control
4450027567



**Mounting
Bolts +
Washers**



Extension Duct
4450017931

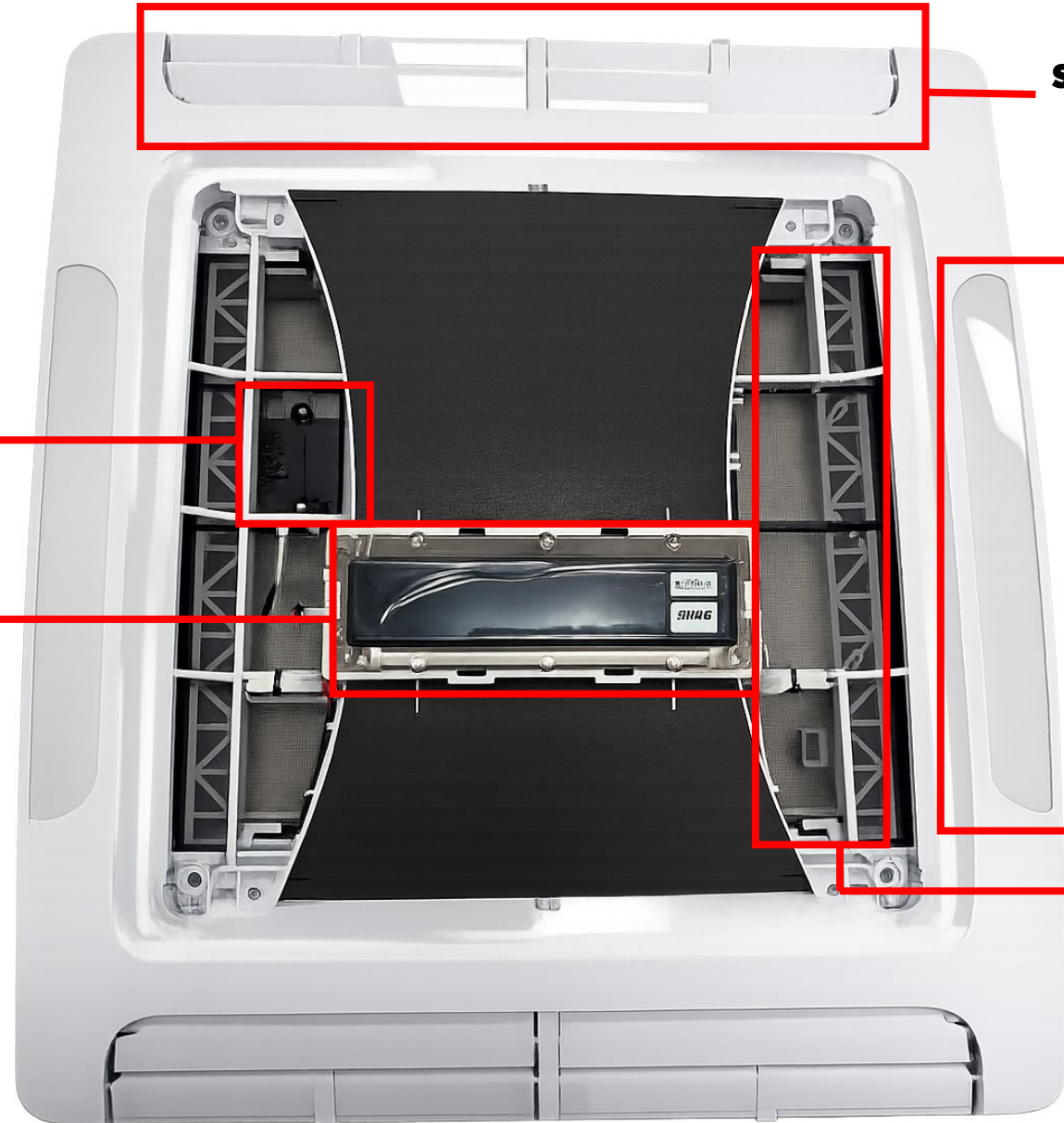
Ionizer

Display PCB
4472004021

Supply Air Louvers
4450032894

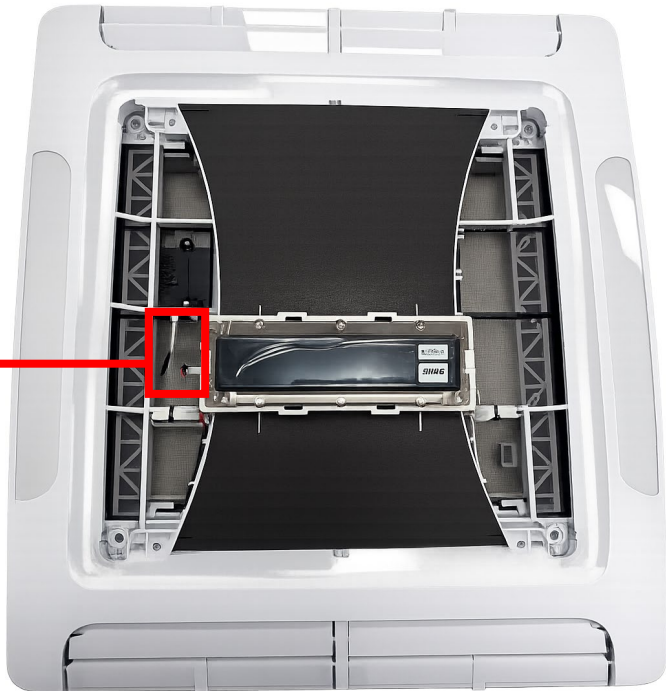
Lights x2
4450029488

Filters x2
4450025683

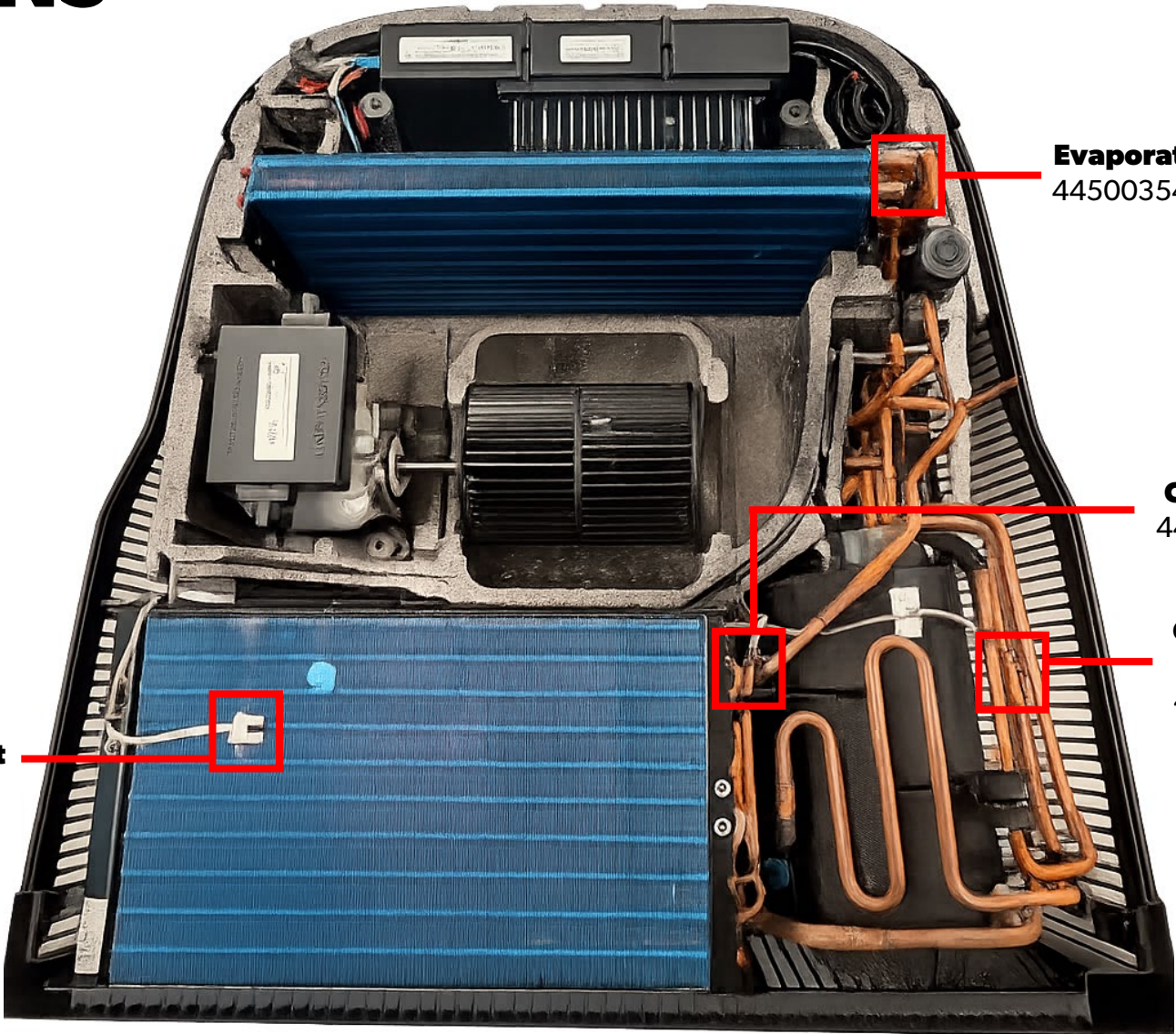


SENSOR LOCATIONS

Indoor Air
4450031469



Outdoor Ambient
4450035403



Evaporator
4450035403

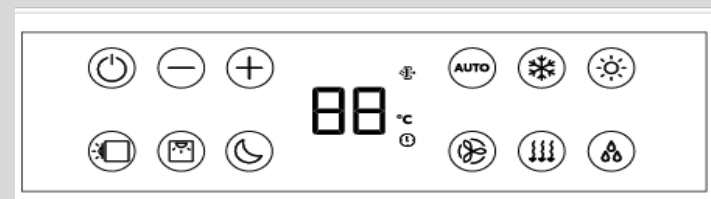
Condenser
4450035403

**Compressor
Discharge**
4450035403

Symbol	Description
	Switches the roof air conditioner on or to standby mode.
	Increases the currently displayed value.
	Reduces the currently displayed value.
	Pairs with bluetooth/WLAN when both buttons are pressed simultaneously.
	Switches the ambient lights in the rooftop unit on or off.
	Switches the ambient lights in the air distribution box on or off.
	Short press: Activates the sleep function (only for models with sleep function). Long press: Press and hold the button for 5 s to enable or disable the beeping sound (only for models with sleep function).
	Displays the set temperature and indicates fan level when fan is adjusted.
	Indicates the Air Purifier mode is active.
	Indicates the temperature is displayed in degrees Celsius.
	Indicates the timer is set.
	Enables/disables the automatic mode (only for models with automatic mode).
	Enables/disables the cooling mode (only for models with cooling mode).
	Enables/disables the heating mode (only for models with heating mode).
	Adjusts the fan speed between Low, Med, High, Turbo or Auto (shown as 1, 2, 3, 4 or AA on the air distribution box display for a short time when changed).
	Enables/disables ventilation mode (only for models with ventilation mode). Circulates air without heating or cooling.
	Enables/disables dehumidification mode (only for models with dehumidification mode).

SYMBOL EXPLANATION

Display



SYMBOL EXPLANATION

Remote Control



Button	Description
—	Display Displays the following: • Time • Selected room temperature (target value) in °C • Actual room temperature • Air conditioning mode • Fan speed • Symbols for active additional functions • Battery status
	Switches the roof air conditioner on or to stand-by mode.
	Increases the currently displayed value.
	Reduces the currently displayed value.
	Selects the fan speed (Selecting the fan speed manually on page 11).
	Selects the air conditioning mode (only for models with automatic mode). The corresponding symbol is displayed.
	Activates the sleep function (only for models with sleep mode). The symbol is displayed.
	Confirms and saves settings.
	Sets the timer and the clock.
	Switches the ambient lights in the air distribution box on or off.
	Press once to display the current interior temperature. The symbol is displayed. Press twice to display the set temperature.
	Activates the I feel function. The symbol is displayed. Every 10 min the I feel function adjusts the target temperature to the actual temperature measured by the remote control.
	Switches the ambient lights in the roof top unit on or off.
	Displays that the remote control's batteries are empty. Replace the batteries (Replacing the remote control batteries on page 14).
	Displays that the remote control is transmitting values to the roof air conditioner.

FAULT FINDING P- ERROR CODES

Protection or Fault Type	Display Code	Cause of Failure	Recommended Actions
Under voltage protection	P1	Input voltage is below 130V ±5V	Check the voltage supply to the unit.
Unit AC over-current protection	P2	Operating AC current exceeds the limit value	1. Check the voltage supply to the unit. 2. Check the condenser and/or evaporator fan motors work. 3. Check the current draw of the compressor when in operation. 4. If it is abnormal, replace the compressor. 5. If OK, replace Main-PCBA.
Discharge tube overheat protection	P4	The temperature of the compressor discharge tube exceeds the protection value.	1. Check the temperature of the discharge tube via the ADB. 2. If it is abnormal, replace the sensor. 3. If it is normal, check the running temperatures for coils. 4. If they are abnormal, check the refrigerant level in the system. 5. If level is abnormal, check for leaks and repair and recharge to correct level.
Evaporator Coil Protection (cooling mode)	P5	Evaporator coil temperature is too low in cooling mode.	1. Check the evaporator fan motor works. 2. Check the temperature of the evaporator coil via the ADB. 3. If it is abnormal, replace the sensor. 4. If it is normal, check the running temperatures for coils. 5. If they are abnormal, check the refrigerant level in the system. 6. If level is abnormal, check for leaks and repair and recharge to correct level.
Condenser Coil Protection (cooling mode)	P6	Condenser coil temperature is too high in cooling mode.	1. Check for any blockages or restrictions of air inlet or outlets to condenser. 2. Check the condenser fan motor works. 3. Check the temperature of the condenser coil via the ADB. 4. If it is abnormal, replace the sensor. 5. If it is normal, check the running temperatures for coils. 6. If they are abnormal, check the refrigerant level in the system. 7. If level is abnormal, check for leaks and repair and recharge to correct level.
Evaporator Coil Protection (heating mode)	P7	Evaporator coil temperature is too high in heating mode.	1. Check the evaporator fan motor works. 2. Check the temperature of the evaporator coil via the ADB. 3. If it is abnormal, replace the sensor. 4. If it is normal, check the running temperatures for coils. 5. If they are abnormal, check the refrigerant level in the system. 6. If level is abnormal, check for leaks and repair and recharge to correct level.
Outdoor Temperature Protection	P8	Outdoor environment temperature under 0°C in Cooling mode or over 32°C in Heating mode.	1. Check if the unit was accidentally switched into Heating Mode in hot ambient or Cooling in cold ambient. 2. If it was definitely in Cooling Mode, check the location of the external sensor is correct. 3. If it has been mis-placed, re-fix it (it could come in contact with something hot like the compressor body or pipework and cause the error). 4. If it is located correctly, then check the temperature of the Outdoor Sensor via the ADB. 5. If all OK, then replace the Main-PCB.
Compressor drive abnormal or compressor doesn't start	P9	Compressor or Main-PCB faulty	If this error occurs, contact Dometic. Replace Main-PCBA or compressor.
IPM module self over temperature or over current	P0	IPM itself faulty or bad soldering	If this error occurs, contact Dometic. Replace Main-PCBA.

FAULT FINDING E- ERROR CODES

Protection or Fault Type	Display Code	Cause of Failure	Recommended Actions
Communication failure	E0	Abnormal communication or outdoor power supply issue inside or outside	1. Check the connection of the communication cable at the Display-PCBA and Main-PCBA. 2. If loose or damaged, repair or replace. 3. If OK, connect a spare Display-PCBA and test. 4. If still abnormal, replace Main-PCBA.
Room temperature sensor fault	E1	Short circuit or open circuit of indoor ambient temperature sensor	1. Check the temperature of the Room Sensor via the ADB. 2. If abnormal, replace the sensor.
Evaporator coil temperature sensor fault	E2	Short circuit or open circuit of evaporator coil temperature sensor	1. Check the temperature of the Evaporator Coil via the ADB. 2. If abnormal, replace the sensor.
Condenser coil temperature sensor fault	E3	Short circuit or open circuit of condenser coil temperature sensor	1. Check the temperature of the Condenser Coil via the ADB. 2. If abnormal, replace the sensor.
LIN-BUS/CI-BUS communication failure	EL	LIN-BUS/CI-BUS communication failure during self-check	1. Check LIN-BUS/CI-BUS connection cable for poor connection or damage. 2. If loose or damaged, repair or replace. 3. Check master control panel working and connection is good. 4. If error still present, replace Display-PCBA.
Indoor Fan Failure	E6	Indoor fan does not rotate or abnormal rotation speed	1. Check if there are any foreign objects stuck and remove them. 2. Check if the connection terminals are loose. If so, re-fix the connection terminals. 3. If no issues are found above, contact Dometic.
Outdoor temperature sensor failure	E7	Outdoor ambient temperature sensor short circuit or open circuit	1. Check the temperature of the Outdoor Ambient Sensor via the ADB. 2. If abnormal, replace the sensor.
Discharge tube temperature sensor failure	E8	Compressor discharge tube temperature sensor short circuit or open circuit	1. Check the temperature of the Compressor Discharge Tube Sensor via the ADB. 2. If abnormal, replace the sensor.
Compressor drive or IPM module fault	E9	Driver or module protection appears 6 times within 30 minutes	1. Check the voltage supply to the unit. 2. Check the condenser and/or evaporator fan motors work. 3. Replace Main-PCBA.
Fault of current sensor	EA	No current was sampled by the system	1. When the code displays immediately after turning the system on (hardware fault), replace the Main-PCBA. 2. If code displays after the compressor starts (compressor current draw is lower than 1A and detection time is more than 30s when compressor works at equal or greater than 35Hz), check refrigerant level in the system. 3. If level is abnormal, check for leaks, repair, and recharge to correct level.
Outdoor EEPROM fault	EE	Unable to read EEPROM data	Replace Main-PCBA.
Outdoor Fan Failure	EF	Outdoor DC fan motor does not rotate or rotates at an abnormal speed	1. Check if there are any foreign objects stuck and remove them. 2. Check if the connection terminals are loose. If so, re-fix the connection terminals. 3. If no issues are found above, contact Dometic.
N/A	EU	Voltage detection error	If this error occurs, contact Dometic.

In previous error code lists, you’ll notice that the instruction “If this error occurs, contact Dometic” appears across multiple error entries.

N/A	EF	If this error occurs, contact Dometic.
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TEMPERATURE SENSOR CHECK

Freshjet 4 & 7 Series:

Item	Sensor	Button Press & Hold Combination	Displayed Indicator
1	Indoor Room Air Temperature	 AND 	
2	Evaporator Coil Temperature	 AND 	
3	Condenser Coil Temperature	 AND 	
4	Compressor Discharge Tube Temperature	 AND 	
5	Outdoor Ambient Temperature	 AND 	

Press and HOLD the button combinations as shown to have the sensor temperature read out on the ADB display.
Once released, the display will return to show the set temperature.

CABLE AND FUSE SIZING

Correct cable and fuse sizing is critical when installing the 48 V Freshjet air-conditioner.
As the unit has a current rating of 28.6 A, 40 Amp wiring and fusing is recommended to ensure safe and reliable operation.

To avoid cables overheating and voltage drop please follow the below table

Amps (A)	Up to 9m (Cable length)	Up to 19m (Cable length)
40 A (Fuse size)	8.4 mm ² (8 AWG) (Cable size)	13.3 mm ² (6 AWG) (Cable size)

Please note that fuses are installed to protect the cabling not the appliance

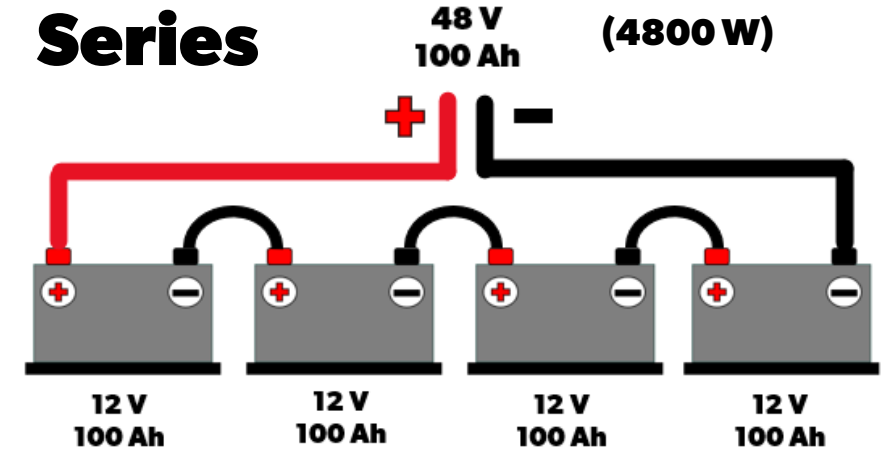
BATTERY CONFIGURATION

Series

Connecting four 12 V 100 Ah batteries in series increases the battery bank voltage to 48 V while maintaining the available capacity at 100 Ah. This is achieved by connecting the positive terminal of each battery to the negative terminal of the next battery, causing the individual battery voltages to add together.

In a series battery bank, the maximum current output at any given moment is limited to the rating of a single battery because the same current must flow through every battery in the series string. Increasing the number of batteries in series increases voltage, but it does not increase the maximum current capability.

Series

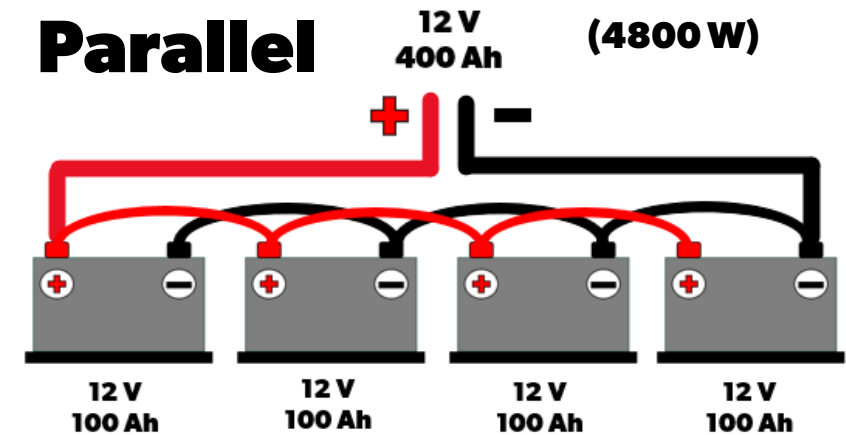


Parallel

Connecting four 12 V 100 Ah batteries in parallel keeps the battery bank voltage at 12 V while increasing the available capacity to 400 Ah. This is achieved by connecting all positive terminals together and all negative terminals together.

In a parallel battery bank, the maximum current output at any given moment increases because each battery shares the load. The total current capability is the sum of the current capability of all batteries connected in parallel.

Parallel



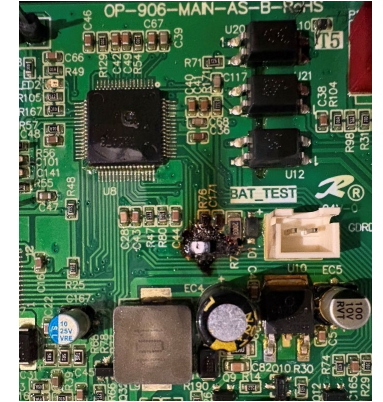
ELECTRICAL CONNECTIONS

Fault

A faulty/poor 48v electrical connection can cause faults within the main PCB.

In this particular case a diode burnt out and triggered EF error.

Outdoor Fan Failure	EF	Outdoor DC fan motor does not rotate or abnormal rotation speed
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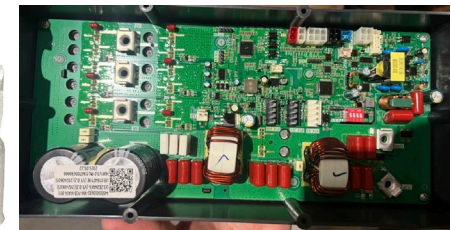
Cause

The negative lug was installed upside down in the supply side Anderson plug, causing the connection to arc. (other electrical connections to check are the shunt, fuse block and battery terminals.)



Remedy

Replace faulty Anderson plug and replace damaged Main PCB



POOR INSTALLATION

ELECTRICAL CONNECTIONS

Fault

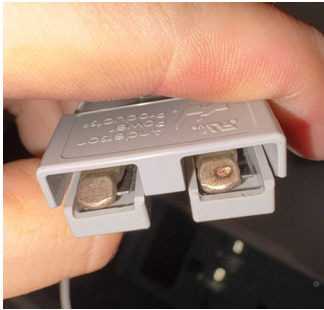
Unit was intermittently faulting on error P5, P7 and P8.

Evaporator Coil Protection (cooling mode)	P5	Evaporator coil temperature is too low in cooling mode
Evaporator Coil Protection (heating mode)	P7	Evaporator coil temperature is too high in heating mode
Outdoor Temperature Protection	P8	Outdoor environment temperature under 0°C in Cooling mode or over 32°C in Heating mode

Cause

Both negative lugs on the Anderson plugs were off centered in opposite directions and showed signs of arcing.

Unit Side



Supply Side



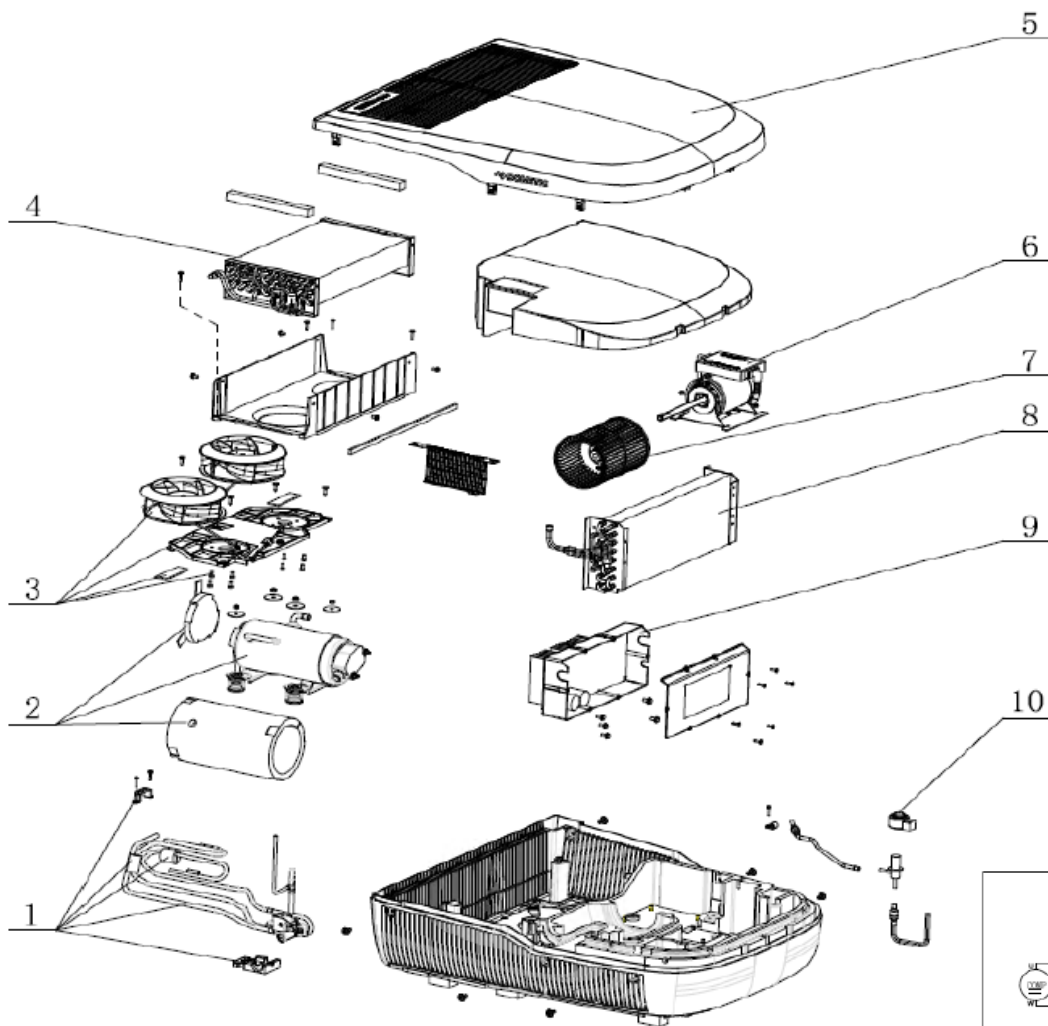
Remedy

Change neutral lug and ensure good connection between plugs.

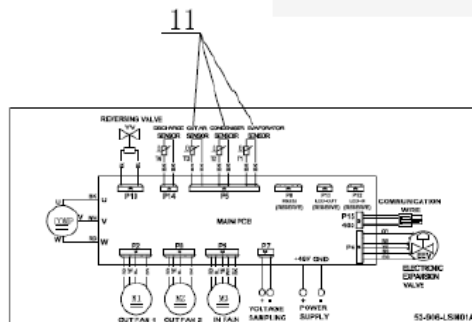


FJX 48V - PARTS BREAKDOWN

9620018598

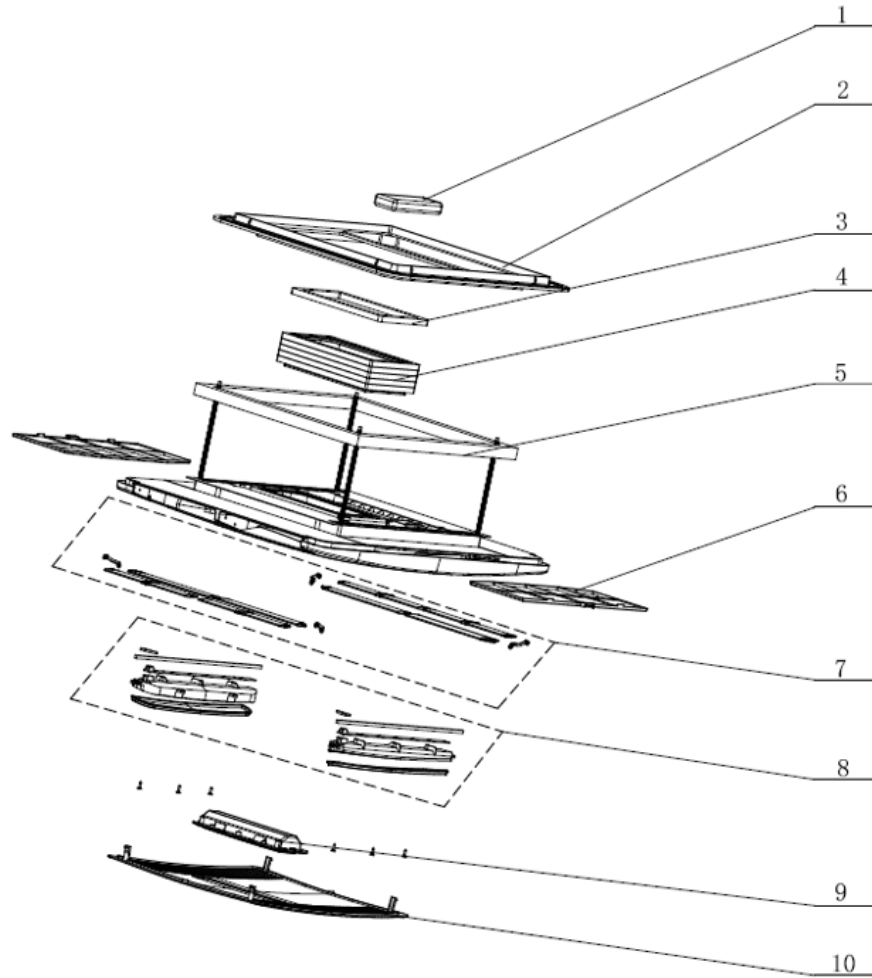


Position	No.	Description
1	4450035402	VALVE REV & PIPES FJ7 LITE 48V
2	4450035399	COMPRESSOR FJ7 LITE 48V
3	4450035400	FAN CONDENSER FJ7 LITE 48V
4	4450033584	CONDENSER FJ7 LITE
5	4472002905	SHROUD FJ7 LITE 48V GREY
6	4450035401	MOTOR FAN EVAP FJ7 LITE 48V
7	4450017916	BLADE FAN IBIS4/LITE/FJ7
8	4450033588	EVAPORATOR FJ7 LITE
9	4450036325	PCB MAIN FJ7 LITE 48V
10	4450033746	VALVE ELECT EXPAN FJ7 LITE 48V
••	4472003967	VALVE & SOLENOID EXPAN FJ7 LITE
11	4450035403	SENSORS SET FJ7 LITE 48V



FJX ADB - PARTS BREAKDOWN

9620001693



Position	Item No	Description
1	4450027567	CONTROL REMOTE FJ4/7
2	4450020818	FRAME MOUNTING FJ4/7
3	4450024602	DUCT UPPER FJ7
4	4450017931	EXTENSION DUCT IBIS4/HARR/FJ
5	4450019320	FOAM INT SQ ADB FJ4/FJ7
6	4450025683	FILTER FJ/IBIS4/PLUS/LITE
7	4450032894	LOUVRE SET ADB FJ4/7 (4)
8	4450029488	LIGHT ASSEMBLY ADB FJ4/7 (2)
9	4472009339	PCB DISPLAY ADB FJX/FJZ
10	4450030829	GRILL RETURN ADB FJ4/7

END

